MIL-C-43035A (GL)

25 August 1965
SUPERSEDING
MIL-C-43035(QMC)
21 August 1961

MILITARY SPECIFICATION

CHAIR, STRAIGHT, KITCHEN

1. SCOPE

1.1 This specification covers one type and size kitchen chair having an upholstered, padded seat and a wire grid back (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces For Organic Coatings.
CCC-A-700 - Artificial Leather, Cloth, Coated, Vinyl Resin, (Upholstery).
PPP-B-636 - Box, Fiberboard.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specification and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

/FSC 7105/

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF COMMERCE

Commercial Standard CS35-61 - Hardwood Plywood

Commercial Standard CS45-60 - Douglas Fir Plywood

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402.)

3. REQUIREMENTS

3.1 Preproduction sample.- When specified (see 6.2), before production is commenced, a sample chair shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- The chair shall be a standard, current product of the manufacturer, except as modified herein. All chairs furnished under any one contract shall be identical.

3.3 Materials.- Materials shall conform to the referenced specifications and requirements specified herein. Where materials are not definitely specified, they shall be of the quality normally used for the purpose in commercial practice.

3.3.1 Plywood.- Plywood shall conform to either of the following:

(a) Douglas Fir, interior type, grade B-D, sanded two sides, conforming to Standard CS45-60.

(b) Hardwood, sanded two sides, constructed of the species and grades of veneers, crossbands, and cores used in commercial practice, minimum of grade 3 face and back veneers, conforming to type II of Standard CS35-61.

3.3.2 Upholstery, artificial leather.- The artificial leather upholstery shall conform to class 4, treatments a-1 (regular fire resistant) and b (mildew resistant); or a-1, b and c (oil resistant) of CCC-A-700. Color and grain shall be "Mottled Sand" in accordance with the approved standard sample (see 6.3).

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3.3.3 Hardware.- All miscellaneous hardware such as screws, bolts, nuts, angle brackets and clips shall be commercial quality, cadmium or zinc plated steel. In lieu of plating, steel clips and angle brackets may be finished according to 3.5.1 and 3.5.2.

3.3.4 Glides.- The glides shall be of the swivel, self leveling type and have floor contact surfaces of stainless steel, molded polyethylene, polyvinyl chloride acetate, or cellulose acetate butyrate.

3.4 Construction.- Construction of the chair shall be in accordance with the manufacturer's commercial practice except for any modifications specified herein. The chair shall consist essentially of a tubular steel frame having four legs; a padded, upholstered seat; and an open wire grid back made up of horizontal and vertical members. There shall be no failure of welds, fracture of metal components, or loosening of fastenings such as screws, spring clips or rivets when the chair is tested as specified in 4.4.4.

3.4.1 Frame.- The frame of the chair shall be fabricated of round section, tubular steel and shall be sturdily constructed and joined (i.e., without wobble). Legs of the chair shall be fabricated by either the single tube or double tube construction methods specified in 3.4.1.1 and 3.4.1.2 respectively. Legs may be tapered or swaged to a smaller diameter as design requires. Legs shall be fitted with glides specified in 3.3.4 and with brass or brass colored anodized aluminum ferrules.

3.4.1.1 Single tube method.- Single tube legs shall be fabricated of not less than 7/8 inch nor more than 1-1/8 inches outside diameter tubular steel with a nominal wall thickness of 0.049 inch. When plywood seat base specified in 3.4.2.1.1 is used, the tubes shall be indented to receive the screw heads. When metal seat base specified in 3.4.2.1.2 is used, legs shall be slotted to receive indents of the spring clips

3.4.1.2 Double tube method.- Double tube legs shall be fabricated of not less than 5/8 inch outside diameter tubular steel with a nominal wall thickness of 0.065 inch. When plywood seat base specified in 3.4.2.1.1 is used, a minimum of two, formed steel clips shall be welded to the front and to each side frame member.. Heavy wire clips shall be inserted through holes in the back frame member. When metal seat base specified in 3.4.2.1.2 is used, legs shall be prepared as specified in 3.4.1.1.

3.4.2 Seat.- Seat of the chair shall consist of a base, padding and upholstery.

3.4.2.1 Base.- Seat base shall be provided with ventilation holes and shall conform to either 3.4.2.1.1 or 3.4.2.1.2.

3.4.2.1.1 Plywood.- Plywood seat base shall be fabricated of 3/4 inch thick plywood specified in 3.3.1. When single tube legs specified in 3.4.1.1 are used,

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the seat base shall be fastened to the leg frame with a minimum of four self tapping screws. When double tube legs specified in 3.4.1.2 are used, the seat base shall be fastened with self tapping screws to the steel clips of the front and side frame members and to the wire clips of the back frame member. At the option of the supplier, machine screws and T-nuts may be used in lieu of tapping screws.

3.4.2.1.2 Metal.- Metal seat base shall be fabricated of steel having a nominal thickness of 0.0359 inch, shaped in accordance with industry's practice. The underside of the seat base shall be fitted with four U-shaped spring clips provided with indents for attachment to the leg frame. The clips shall be riveted to the seat base.

3.4.2.2 Padding.- Seat padding shall be fabricated of either a 2 inch minimum thickness of molded pure vinyl foam or a 1½ inch minimum thickness of sheet foam rubber with ¼ inch minimum thickness of blended cotton felt topping.

3.4.2.3 Upholstery.- Seat upholstery shall be fabricated of material specified in 3.3.2. The upholstery shall be applied tightly over the padding specified in 3.4.2.2 with no wrinkles and fastened to the underside of the seat base.

3.4.3 Back.- Back of the chair shall be fabricated of a wire grid inclosed within the tubular steel frame. The wire shall be inserted into holes at the top of the tubular steel frame and attached to the underside of the seat base. The wire grid members shall be not less than 0.162 inch nominal diameter steel wire, arranged as vertical and horizontal grid members, and shall be welded at the intersections of the wire.

3.4.4 Welding.- All welds shall be smooth, sound, continuous, and shall have all excess flux removed prior to finishing.

3.4.5 Driving of screws and rivets.- Screws and rivets shall be thoroughly driven with heads flush against surfaces. All rivets shall be securely and neatly set by heading, peening or upsetting the ends.

3.5 Finish.-

3.5.1 Ferrous surface preparation.- Exposed ferrous surfaces shall be cleaned and prepared for painting in accordance with type I or II of TT-C-490.

3.5.2 Painting.- Ferrous surfaces, pre-treated as specified in 3.5.1, shall be coated with a commercial bronze enamel. The paint film shall be smooth, uniform in color, and free from sags, runs, creep, orange peel, pits, streaks and areas of corrosion or no film.

3.5.2.1 Adhesion, toughness, and flexibility of paint film.- When subjected to the knife test specified in 4.4.1, it shall be difficult to furrow through the film, indicating adhesion and toughness of the film. The edge of the furrow shall be free from the jagged appearance indicative of brittleness of the coating.

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3.5.2.2 Alkali resistance of paint film.- When subjected to the alkali test specified in 4.4.2, there shall be no wrinkling, blistering or other failure of the film immersed. After a 24-hour drying period, there shall be no more than a slight difference in color between the portion that was immersed and the portion that was not immersed.

3.5.2.3 Salt spray resistance of paint film.- When subjected to the salt spray test specified in 4.4.3, there shall be no wrinkling, blistering, or other visible failure of the film. After a 24-hour waiting period following the test, the film on the exposed sample shall be as hard as the film on the comparison, unexposed sample.

3.6 Marking for identification.- The underside of the chair seat shall be permanently and distinctly stamped or marked for identification. The letters "US" shall be in characters 1-inch high and the name of the manufacturer or brand or mark of such known character as to make the source of manufacture easily determined, nomenclature, number and year of contract shall be in characters from ¼-inch to ½-inch high.

3.7 Workmanship.- The finished chair shall be clean and free from any rough or sharp edges, burrs or slivers, oil, grease and dirt. The painted surfaces shall be free from nicks, gouges, scratches or abrasions noticeable at a distance not less than five feet.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection.- When a preproduction sample is required, it shall be examined for defects in table III, dimensions specified, and tested as specified in 4.4.4.

4.3 Sampling for inspection and acceptance.- Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 In-process inspection.-

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4.3.2.1 In-process examination.- Examination shall be made during the manufacturing process for the requirements specified in table I to establish that no deviation is made from the indicated requirements. Whenever nonconformance is noted, correction shall be made to affected items and process.

TABLE I.- In-process examination

Requirement	Requirement reference
Welding	3.4.1.2, 3.4.3, and 3.4.4
Cleaning and pretreatment	3.5.1

4.3.2.2 Paint film testing.- Testing shall be performed in accordance with table II for the characteristics shown therein. At the same time that ferrous surfaces are cleaned, prepared for painting and painted with enamel, three (3) test panels, measuring 3-inches by 6-inches, shall be prepared concurrent with production. These panels shall be prepared on each lot or batch of finishing materials or when any change occurs in the application of finishing materials. In lieu of flat test panels, it will be permissible to use painted tubular chair components, or sections thereof at least 6 inches long.

TABLE II.- Paint film testing

Characteristic	Requirement paragraph	Test method paragraph	Sample unit	Results reported as
Adhesion, toughness and flexibility	3.5.2.1	4.4.1	1 test panel (*)	Pass or fail
Alkali resistance	3.5.2.2	4.4.2	1 test panel (*)	Pass or fail
Salt spray resistance	3.5.2.3	4.4.3	1 test panel (*)	Pass or fail

(*) For each lot or batch of finishing materials or for each change in application of finishing materials.

4.3.3 End item inspection.- The sample unit for examination shall be one completely fabricated chair. The lot shall be all the units submitted for inspection at one time.

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4.3.3.1 Visual examination.- Examination shall be made of the end item for defects listed in table III. The inspection level shall be II with an acceptable quality level (AQL) of 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE III.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish			
Ferrous surfaces	Not bronze color.	X	
	Film not smooth or color is not uniform.		X
	Sag, run, creep, orange peel, pit, streak, or area of corrosion or no film.		X
	Nicks, gouges, scratches or abrasions noticeable at a distance not less than five feet.		X
Ferrule	Not brass color.		X
General	Not clean.		X
	Rough edge.		X
	Sharp edge, burr or sliver.	X	
	Evidence of oil, grease or dirt.		X
Construction and workmanship	Frame not sturdily constructed and joined, i.e., wobbles.	X	
	Leg not single or double tube construction.	X	
	Leg not fitted with glide and ferrule.	X	
	When plywood seat base is used:		
	Leg tube not indented to receive the screw head (applicable to single tube construction).		X
	Steel (welded) clip missing (applicable to double tube constructions).		X
	Screw not type specified.		X
	When metal seat base is used:		
	Leg not slotted to receive indents of the spring clip (applicable to single tube construction).		X
	U-shaped spring clip missing, not provided with indent, or not assembled with the leg frame.		X
	Ventilation holes in seat base missing.		X
	Wire clip missing or not fastening the seat base to the back frame member (applicable to double tube construction).		X

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TABLE III.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (cont'd)	Padding missing.	X	
Upholstery	Missing.	X	
	Not applied as specified, i.e., has wrinkle or is loose on seat.	X	
	Not fastened as specified.		X
Back	Not fabricated as specified, i.e., not wire grid.		X
	Wire grid not inclosed within and inserted into holes at the top of the tubular frame and attached to the underside of the seat base.		X
	Grid members are not arranged as specified. i.e., vertical and horizontal.		X
Driving of screws and rivets	Screw or rivet missing, broken, damaged, or not thoroughly driven with head flush against surface.		X
	U-shaped spring clip not riveted to seat base (applicable to metal seat base).		X
	Rivet not securely and neatly set but cannot be removed by hand.		X
Marking	Missing, illegible, misplaced, incomplete, or misspelled.		X

4.3.3.2 Dimensional examination.- Examination shall be made of the chair for defects in dimensions. Any dimension not within specified tolerance shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.3.3 Testing of the end item.- Testing of the end item shall be performed in accordance with the test indicated in table IV for the characteristics shown therein. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

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TABLE IV.- Testing of the end item

Characteristics	Requirement paragraph	Test method paragraph	Sample unit	Results reported as
Soundness of welds, strength of metal components, tightness of fastenings	3.4	4.4.4	1 chair	Pass or fail

4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing and markings are in compliance with section 5 of this specification. Defects shall be scored as specified in table V. The lot shall be the number of containers offered for delivery at one time. The sample unit shall be one shipping container fully prepared for delivery except that it need not be sealed. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE V. - Examination of preparation for delivery

Examine	Defect
Marking	Missing, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any nonconforming component, component missing, damaged or otherwise defective.
Content	Content is more or less than required.

4.4 Tests.-

4.4.1 Knife test.- The sample test panel or chair component section shall be subjected to the knife test in accordance with method 6304 of Federal Test Method Standard No. 141. Examination shall then be made to determine compliance with 3.5.2.1.

4.4.2 Alkali resistance test.- A portion of the sample test panel or chair component section shall be immersed in a 5 percent solution of sodium hydroxide at room temperature for 30 minutes. Examination shall be made to determine compliance with the appearance requirements of 3.5.2.2. The sample shall be allowed to dry in air for a period of 24 hours. Examination shall then be made to determine compliance with the color requirement of 3.5.2.2.

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(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(d) If weather-resistant fiberboard shall be required for level B packing (see 5.2.2.1).

6.3 For samples of the approved color and grain of mottled sand referenced in 3.3.2, address the procuring office issuing the invitation for bids.

6.4 Packaging.- The method of packaging referred to in 5.1.1, is similar to method III of MIL-P-116.

6.5 Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

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Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION — CITY AND STATE

CONTRACT NO — QUANTITY OF ITEMS PROCURED — DOLLAR AMOUNT — $

MATERIAL PROCURED UNDER A: ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* — DATE

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